

$$\Delta U = Q_{in} + W_{on} = 0$$

14 The phase difference can be calculated using:

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi f}{v} \Delta x = \frac{2\pi(500)}{340} (0.425) = \frac{5}{4} \pi$$

15 Period of oscillation, $T = 2\pi \sqrt{\frac{m}{k}}$ dependent on mass and spring constant.